University of Illinois
ECE 310 **Profs. Do, Snyder**

Midterm Exam 1

7:00-9:00pm, Wednesday, October 1, 2025

Problem	Pts.	Score
1	12	
2	12	
3	10	
4	10	
5	15	
6	16	
7	15	
8	10	
Total	100	

Name: Solutions

NetID: _____

Instructions

- You may not use any books, calculators, or notes other than one handwritten two-sided sheet of 8.5" x 11" paper.
- Show all your work to receive full credit for your answers.
- When you are asked to "calculate", "determine", or "find", this means providing closed-form expressions (i.e., without summation or integration signs).
- Neatness counts. If we are unable to read your work, we cannot grade it.
- Turn in your entire booklet once you are finished. No extra booklet or papers will be considered.

(12 Pts.)

1. Select **True** or **False** to each of the following statements:

   (a) The impulse response of a given LTI system is known to be a bounded sequence. This system must be BIBO stable. **True**/**False**

   (b) The convolution of two causal signals always results in a causal signal. (A causal signal is defined as a signal that is zero for all negative time values.) **True**/**False**

   (c) A BIBO stable LTI system has a transfer function with two poles at $z = \frac{1}{4}$ and $z = 3$. The impulse response of this system must be two-sided. **True**/**False**

   (d) An LTI system with a left-sided impulse response can never be causal. **True**/**False**

   (e) An unbounded signal is passed as input to a BIBO stable LTI system. The resulting output is always an unbounded signal. **True**/**False**

   (f) The parallel connection of two unstable LTI systems always forms another unstable LTI system. **True**/**False**

(12 Pts.)

2. For each of the systems with input $x[n]$ and output $y[n]$ shown in the table, indicate by "**yes**" or "**no**" whether the properties indicated apply to the system.

	Linear	Shift-Invariant	Causal	Stable
$y[n] = \|n\|x[n]$	Yes	No	Yes	No
$y[n] = \|x[n] - x[n-1]\|$	No	Yes	Yes	Yes
$y[n] = x[n] * 2^n u[-n]$	Yes	Yes	No	Yes

(10 Pts.)

3. For each of the following parts, compute the convolution $y[n] = (x * h)[n]$ between the following given signals.

   (a) $x[n] = \{\underset{\uparrow}{1}, 2, 3, 0, -1, -2, -3\}, \quad h[n] = \{1, 0, \underset{\uparrow}{1}\}$

$y = Hx$

$y[n] = \{1, 2, \underset{\uparrow}{4}, 2, 2, -2, -4, -2, -3\}$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}_{9\times 7} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 0 \\ -1 \\ -2 \\ -3 \end{bmatrix}_{7\times 1} = \begin{bmatrix} 1 \\ 2 \\ 4 \\ 2 \\ 2 \\ -2 \\ -4 \\ -2 \\ -3 \end{bmatrix} \leftarrow n=0$$

   (b) $x[n] = nu[n+1]u[-n+1], \quad h[n] = n\left(\frac{1}{3}\right)^n \cos(n)$

$x[n] = \{-1, \underset{\uparrow}{0}, 1\} = -\delta[n+1] + \delta[n-1]$

$$y[n] = x[n] * h[n]$$
$$= (-\delta[n+1] + \delta[n-1]) * h[n]$$
$$= -h[n+1] + h[n-1]$$

$$y[n] = -(n+1)\left(\frac{1}{3}\right)^{n+1}\cos(n+1) + (n-1)\left(\frac{1}{3}\right)^{n-1}\cos(n-1)$$

(10 Pts.)

4. The moving average system is a discrete-time system where L recent input samples are averaged to produce the system output at each time index for finite, positive integer L. Consider the below modification of the moving average system.

$$y[n] = \frac{1}{L}\sum_{k=0}^{L-1} x[n - Sk]$$

Above, S is a positive integer.

(a) Determine if the above modified moving average system is **(i)** linear and **(ii)** time-invariant. Please ~~circle~~ one option for each part, no proofs or justification necessary.

i. **Linear** **Non-linear**

ii. **Time-invariant** **Time-varying**

(b) Let $L = 3$ and $S = 4$ above. Determine the impulse response $h[n]$ of this system.

$$y[n] = \frac{1}{3}\sum_{k=0}^{2} x[n-4k] = \frac{1}{3}x[n] + \frac{1}{3}x[n-4] + \frac{1}{3}x[n-8]$$

$$h[n] = \frac{1}{3}\delta[n] + \frac{1}{3}\delta[n-4] + \frac{1}{3}\delta[n-8]$$

$$= \left\{\frac{1}{3}, 0, 0, 0, \frac{1}{3}, 0, 0, 0, \frac{1}{3}\right\}$$

(c) Consider the following statement: The moving average system in this problem is always BIBO stable. Circle one of the following and provide a brief explanation.

True/**False**

Explanation: The given system is an LCCDE with no feedback terms. Therefore, the system is FIR and must be BIBO stable, i.e. impulse response is absolutely summable.

(15 Pts.)

5. For each of the following discrete-time signals, determine its z-transform and region of convergence.

(a) $x[n] = e^{j\frac{\pi}{3}n}u[n+4]$

$$X[n] = e^{j\frac{\pi}{3}(-4)}\, e^{j\frac{\pi}{3}(n+4)}\, u[n+4]$$

$$X(z) = \frac{e^{-j\frac{4\pi}{3}} z^4}{1 - e^{j\frac{\pi}{3}n} z^{-1}}, \quad ROC = |z| < \infty$$

(b) $x[n] = u[n] - u[n-8] = \{\underset{\uparrow}{1}, 1, 1, 1, 1, 1, 1, 1, 0, 0, \cdots\}$

$$X(z) = 1 + z^{-1} + z^{-2} + z^{-3} + z^{-4} + z^{-5} + z^{-6} + z^{-7}, \quad ROC = z \neq 0$$

(c) $x[n] = \cos^2\left(\frac{\pi}{4}n\right)u[n]$

$$X[n] = \left(\frac{e^{j\frac{\pi}{4}n} + e^{-j\frac{\pi}{4}n}}{2}\right)^2 u[n]$$

$$= \left(\frac{1}{4}e^{j\frac{\pi}{2}n} + \frac{1}{2} + \frac{1}{4}e^{-j\frac{\pi}{2}n}\right)u[n]$$

$$\updownarrow \mathcal{Z}$$

$$X(z) = \frac{\frac{1}{4}}{1 - e^{j\frac{\pi}{2}}z^{-1}} + \frac{\frac{1}{4}}{1 - e^{-j\frac{\pi}{2}}z^{-1}} + \frac{\frac{1}{2}}{1 - z^{-1}}, \quad ROC = |z| > 1$$

(16 Pts.)

6. Consider the below **causal** LTI system expressed by an LCCDE.

$$y[n] = \frac{4}{3}y[n-1] + \frac{4}{3}y[n-2] + x[n] - x[n-2]$$

(a) Determine the transfer function $H(z)$, poles, zeros, and ROC of the system.

$$Y(z) = \frac{4}{3}z^{-1}Y(z) + \frac{4}{3}z^{-2}Y(z) + X(z) - z^{-2}X(z)$$

$$Y(z)\left(1 - \frac{4}{3}z^{-1} - \frac{4}{3}z^{-2}\right) = X(z)\left(1 - z^{-2}\right)$$

$$\frac{Y(z)}{X(z)} = H(z) = \frac{1 - z^{-2}}{1 - \frac{4}{3}z^{-1} - \frac{4}{3}z^{-2}} = \frac{(1+z^{-1})(1-z^{-1})}{(1-2z^{-1})(1+\frac{2}{3}z^{-1})}, \quad |z| > 2$$

$H(z) =$ $\dfrac{1 - z^{-2}}{1 - \frac{4}{3}z^{-1} - \frac{4}{3}z^{-2}}$ ROC = $|z| > 2$

Poles = $z = 2,\ z = -\frac{2}{3}$ Zeros = $z = 1,\ z = -1$

(b) Determine the system response $y[n]$ to input $x[n] = 3\delta[n] + 2\delta[n-1]$.

$$Y(z) = X(z)H(z) = (3 + 2z^{-1}) \cdot \frac{(1+z^{-1})(1-z^{-1})}{(1-2z^{-1})(1+\frac{2}{3}z^{-1})}$$

$$= 3\left(1 + \frac{2}{3}z^{-1}\right) \cdot \frac{(1+z^{-1})(1-z^{-1})}{(1-2z^{-1})(1+\frac{2}{3}z^{-1})}$$

$$= \frac{3(1-z^{-2})}{1-2z^{-1}} = \frac{3 - 3z^{-2}}{1 - 2z^{-1}}$$

$$y[n] = 3(2)^n u[n] - 3(2)^{n-2} u[n-2]$$

* Alternatively, find $h[n]$, then $y[n] = x[n] * h[n]$

$= (3\delta[n] + 2\delta[n-1]) * h[n]$

$= 3h[n] + 2h[n-1]$

(15 Pts.)

7. The transfer function $H(z)$ of a **BIBO stable** LTI system is given below.

$$H(z) = \frac{1 - z^{-1}}{(1 + 2z^{-1})(1 + \frac{2}{3}z^{-1})}$$

(a) Determine the impulse response $h[n]$ of this LTI system.

$H(z) = \frac{1-z^{-1}}{(1+2z^{-1})(1+\frac{2}{3}z^{-1})} = \frac{A_1}{1+2z^{-1}} + \frac{A_2}{1+\frac{2}{3}z^{-1}} = \frac{\frac{9}{4}}{1+2z^{-1}} + \frac{-\frac{5}{4}}{1+\frac{2}{3}z^{-1}}$, $\frac{2}{3} < |z| < 2$

$1 - z^{-1} = A_1(1+\frac{2}{3}z^{-1}) + A_2(1+2z^{-1})$

A_1: $z = -2$

$1 - (-2)^{-1} = A_1(1+\frac{2}{3}(-2)^{-1})$

$\frac{3}{2} = A_1(\frac{2}{3})$

$A_1 = \frac{9}{4}$

A_2: $z = -\frac{2}{3}$

$1 - (-\frac{2}{3})^{-1} = A_2(1+2(-\frac{2}{3})^{-1})$

$\frac{5}{2} = A_2(-2)$

$A_2 = -\frac{5}{4}$

$h[n] = -\frac{9}{4}(-2)^n u[-n-1] - \frac{5}{4}(-\frac{2}{3})^n u[n]$

$h[n] = -\frac{9}{4}(-2)^n u[-n-1] - \frac{5}{4}(-\frac{2}{3})^n u[n]$

(b) With the above impulse response $h[n]$, this LTI system can be implemented as $H(z) = H_1(z) + H_2(z)$, or $Y(z) = (H_1(z) + H_2(z))X(z) = H_1(z)X(z) + H_2(z)X(z)$. Determine **BIBO stable** difference equations that implement the corresponding systems $H_1(z)$ and $H_2(z)$. Make sure to specify the directions of recursive computation, i.e. causal or anti-causal implementations.

$H_1(z) = \frac{Y_1(z)}{X_1(z)} = \frac{\frac{9}{4}}{1+2z^{-1}}$, $|z| < 2$ (anti-causal)

$Y_1(z)(1+2z^{-1}) = \frac{9}{4}X_1(z)$

$y_1[n] + 2y_1[n-1] = \frac{9}{4}x_1[n]$ (anti-causal)

$y_1[n-1] = -\frac{1}{2}y[n] + \frac{9}{8}x_1[n]$ (backwards in time)

$H_2(z) = \frac{Y_2(z)}{X_2(z)} = \frac{-\frac{5}{4}}{1+\frac{2}{3}z^{-1}}$, $|z| > \frac{2}{3}$ (causal)

$Y_2(z)(1+\frac{2}{3}z^{-1}) = -\frac{5}{4}X_2(z)$

$y_2[n] + \frac{2}{3}y_2[n-1] = -\frac{5}{4}x_2[n]$

$y_2[n] = -\frac{2}{3}y_2[n-1] - \frac{5}{4}x_2[n]$ (causal)

(forward in time)

(10 Pts.)

8. The two parts of this problem are unrelated.

(a) An LTI system has the following transfer function $H(z)$.

$$H(z) = \frac{1 - \frac{3}{4}z^{-1}}{1 + 3z^{-1}}, \ |z| > 3.$$

Mark "**True**" or "**False**" which of the following inputs produce a **bounded output**.

i. $\left(\frac{3}{4}\right)^n u[n]$ **True/False**

ii. $\frac{1}{3}\left(\frac{1}{2}\right)^n u[n] + \left(\frac{1}{2}\right)^{n-1} u[n-1]$ **True/False**

iii. $(-3)^n u[n]$ **True/False**

iv. $\delta[n] - \frac{3}{4}\delta[n-1]$ **True/False**

v. $\delta[n] + 3\delta[n-1]$ **True/False**

(b) A causal LTI system has the following transfer function $H(z)$.

$$H(z) = \frac{1}{(1 - \frac{3}{4}z^{-1})(1 - jz^{-1})(1 - e^{j\frac{2}{3}}z^{-1})}, \ |z| > 1.$$

Mark "**True**" or "**False**" which of the following inputs produce an **unbounded output**.

i. $\delta[n] - \frac{3}{4}\delta[n-1]$ **True/False**

ii. $\left(\frac{2}{3}\right)^n u[n]$ **True/False**

iii. $j^n u[n]$ **True/False**

iv. $\sin\left(\frac{\pi}{2}n\right) u[n]$ **True/False**

v. $\cos\left(\frac{2\pi}{3}n\right) u[n]$ **True/False**

Scratch paper: Do not remove from the exam!